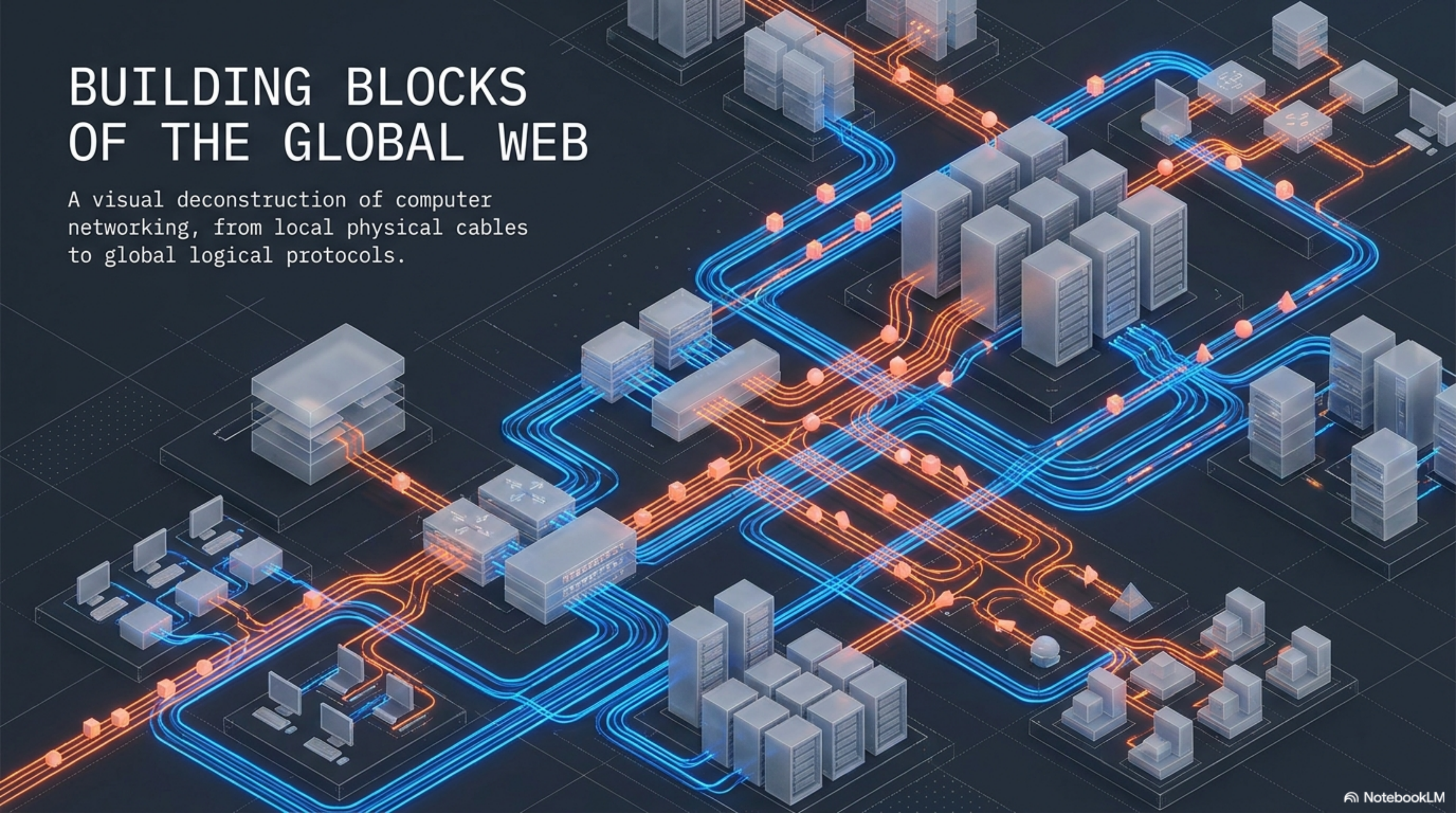


BUILDING BLOCKS OF THE GLOBAL WEB

A visual deconstruction of computer networking, from local physical cables to global logical protocols.



Every Network Begins with **Physical Hosts, Nodes, and Media**

The Endpoints (Hosts)

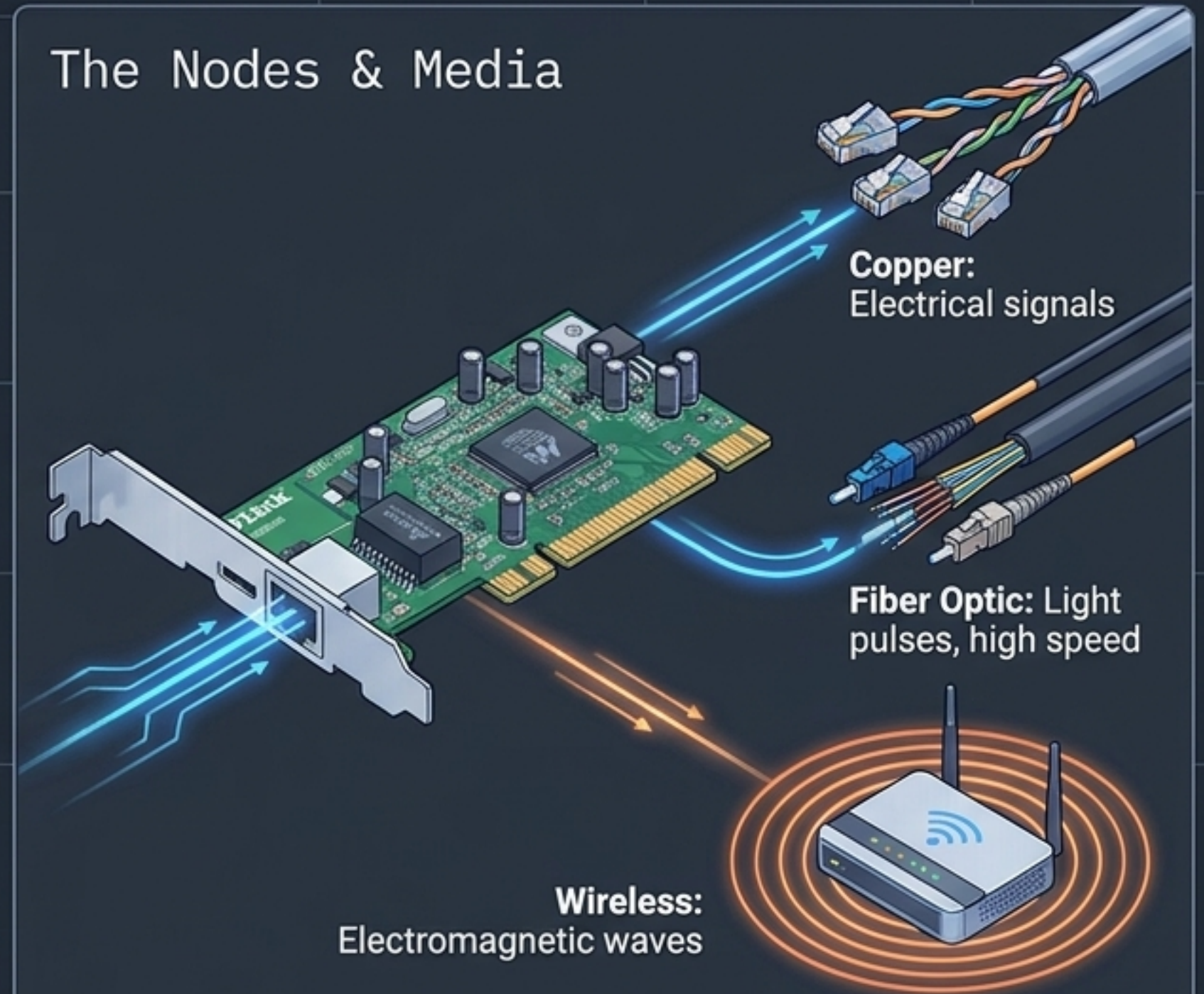


Client: Ordinary workload, initiates requests.

Server: Runs 24/7, handles concurrent connections.

- Xeon processors (multiprocessing)
- ECC RAM (error correction)
- RAID storage (hot-swappable redundancy)

The Nodes & Media



Copper:
Electrical signals

Fiber Optic: Light pulses, high speed

Wireless:
Electromagnetic waves

Networks Scale Logarithmically from Desktops to Continents



PAN (Personal)

~10 meters. Bluetooth, NFC, USB. Linking a smartphone and headphones.



LAN / WLAN (Local)

A single building or home. Ethernet/WiFi.



CAN (Campus)

Multiple buildings/LANs joined in a specific area (e.g., a university).



MAN (Metropolitan)

Spanning a city using fiber-optic backbones.



WAN (Wide)

The largest scale. The Internet itself, connecting global continents.

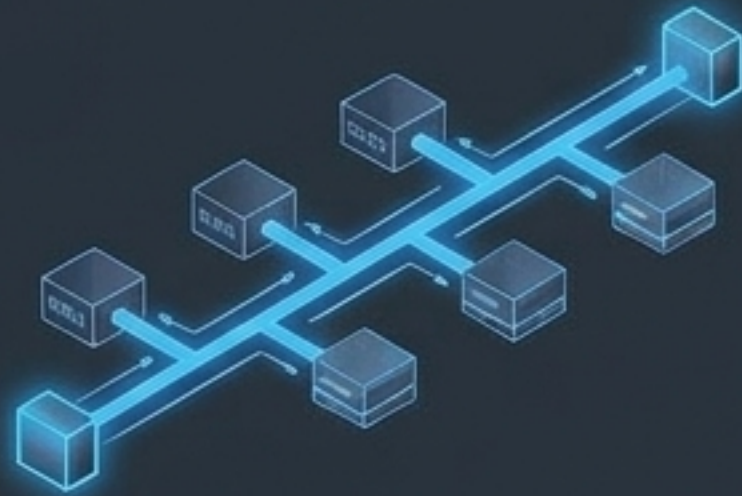


Note on SAN:

Storage Area Network. Partitioned specifically for high-speed data storage arrays, bypassing standard LAN traffic.

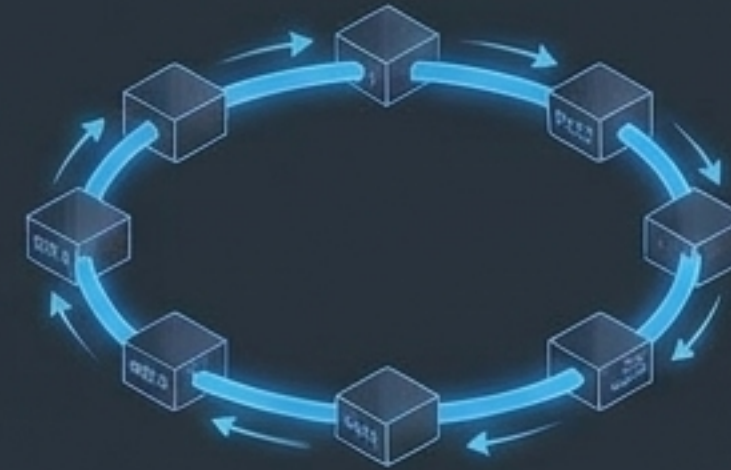
LAN Topologies Dictate the Shape of Local Traffic

Bus Topology



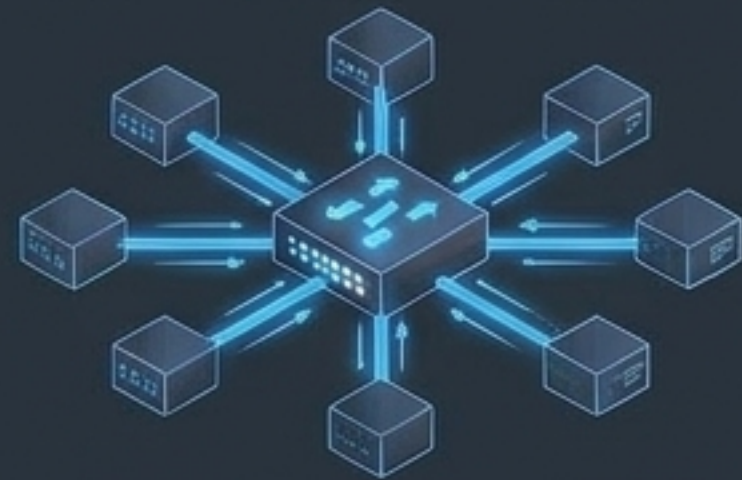
Single shared cable.
Requires terminators at ends to prevent signal reflection.
If the bus breaks, the network collapses.

Ring Topology



Closed loop. Data passes node-to-node.
No terminators needed, but highly vulnerable to single-point failure.

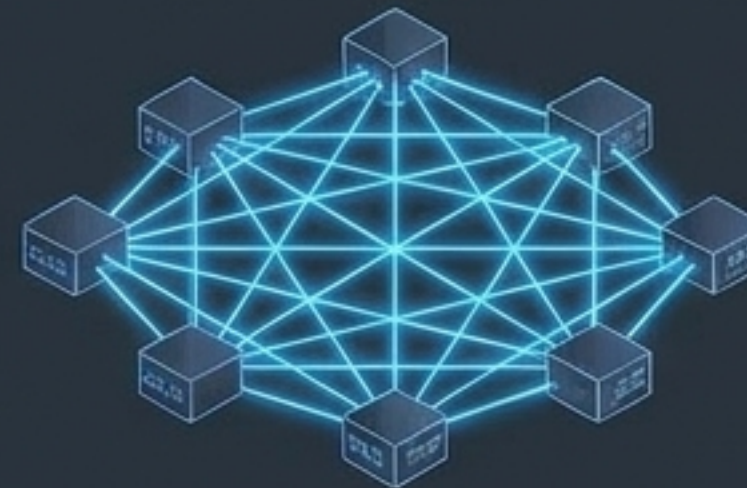
Star Topology



Star Topology

Most common Ethernet LAN.
All nodes connect to a central switch.
High fault tolerance, but the central switch is a single point of failure.

Mesh Topology



Mesh Topology

Every node connects to every other node. Highest fault tolerance.

Fully meshed formula: $C = n(n-1)/2$.
(e.g., 6 devices = 15 connections; 100 devices = 4,950 connections—making it prohibitively expensive for physical wires).

Hardware Intelligence Evolves to Reduce Unnecessary Traffic

Level 3: The Router (Layer 3 - Network)

The Gateway

Connects entirely different networks (e.g., a LAN to a WAN).
Reads IP addresses to reject outside traffic or route internal traffic to the global internet.

Level 2: The Switch (Layer 2 - Data Link)

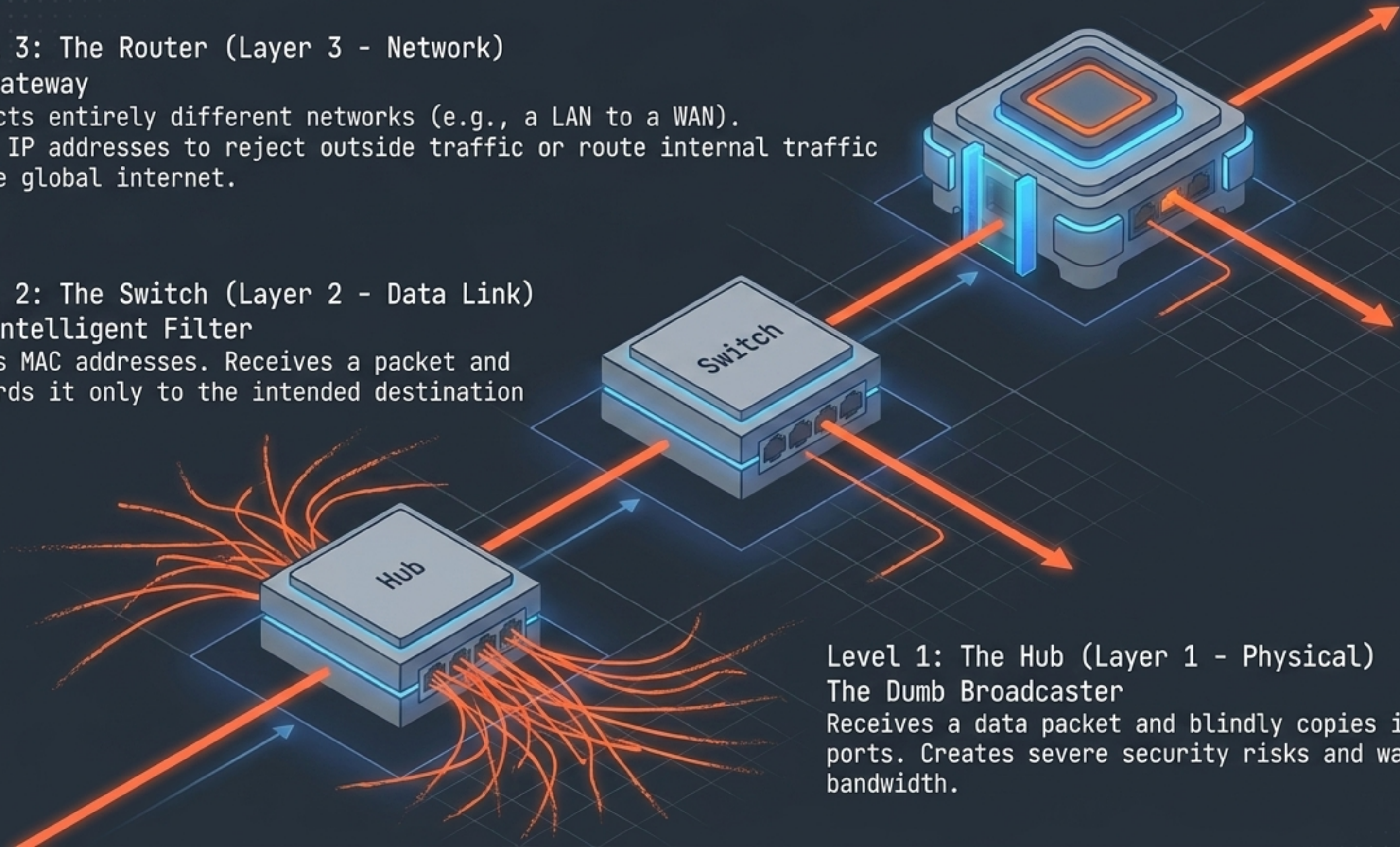
The Intelligent Filter

Learns MAC addresses. Receives a packet and forwards it only to the intended destination port.

Level 1: The Hub (Layer 1 - Physical)

The Dumb Broadcaster

Receives a data packet and blindly copies it to all ports. Creates severe security risks and wastes bandwidth.



The Address Matrix: Physical Identity vs. Logical Location

MAC Address Media Access Control



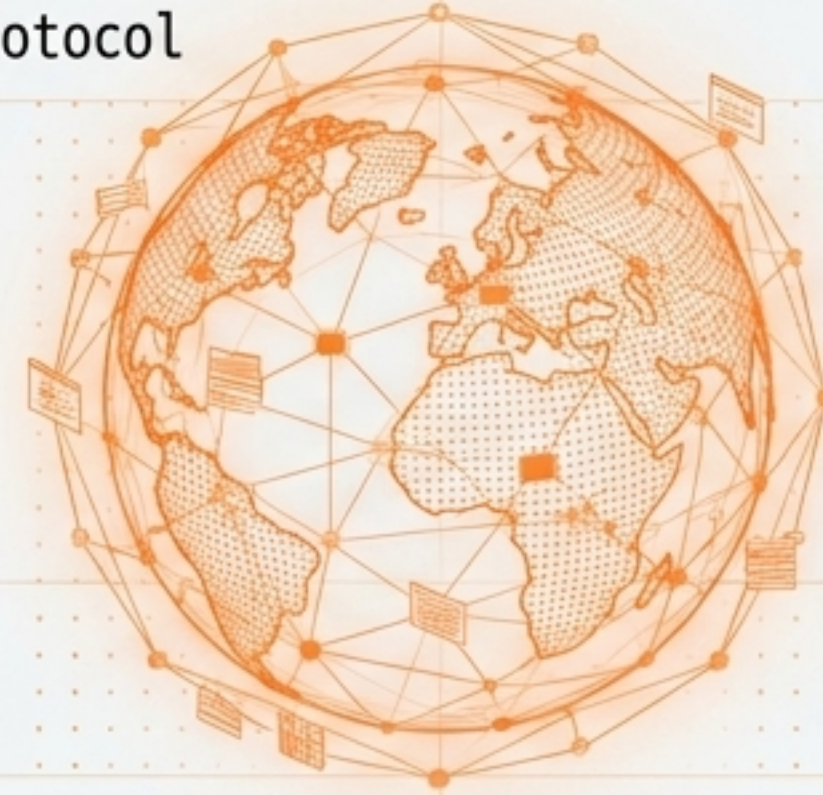
- **Analogy:** “Who you are” (like a name).

Nature: Static. Burned into the hardware NIC by the manufacturer. Never changes.

Format: 48-bit hexadecimal (e.g., 00-16-EA-06-6C-3E). Allows for $\approx 2.8 \times 10^{14}$ unique addresses.

- **Scope:** Used by Switches for local (LAN) delivery.

IP Address Internet Protocol



- **Analogy:** “Where you are” (like a mailing address).

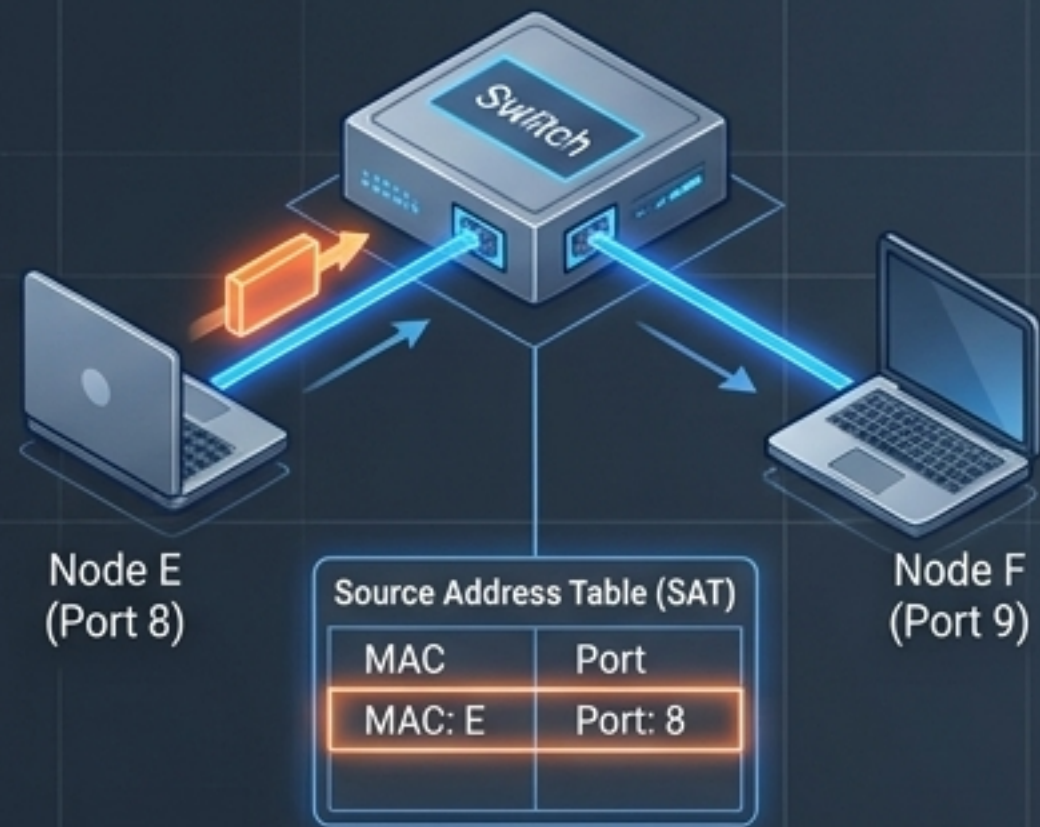
Nature: Dynamic. Assigned by the network. Changes when you move to a new location.

Format: IPv4 (32-bit) or IPv6 (128-bit).

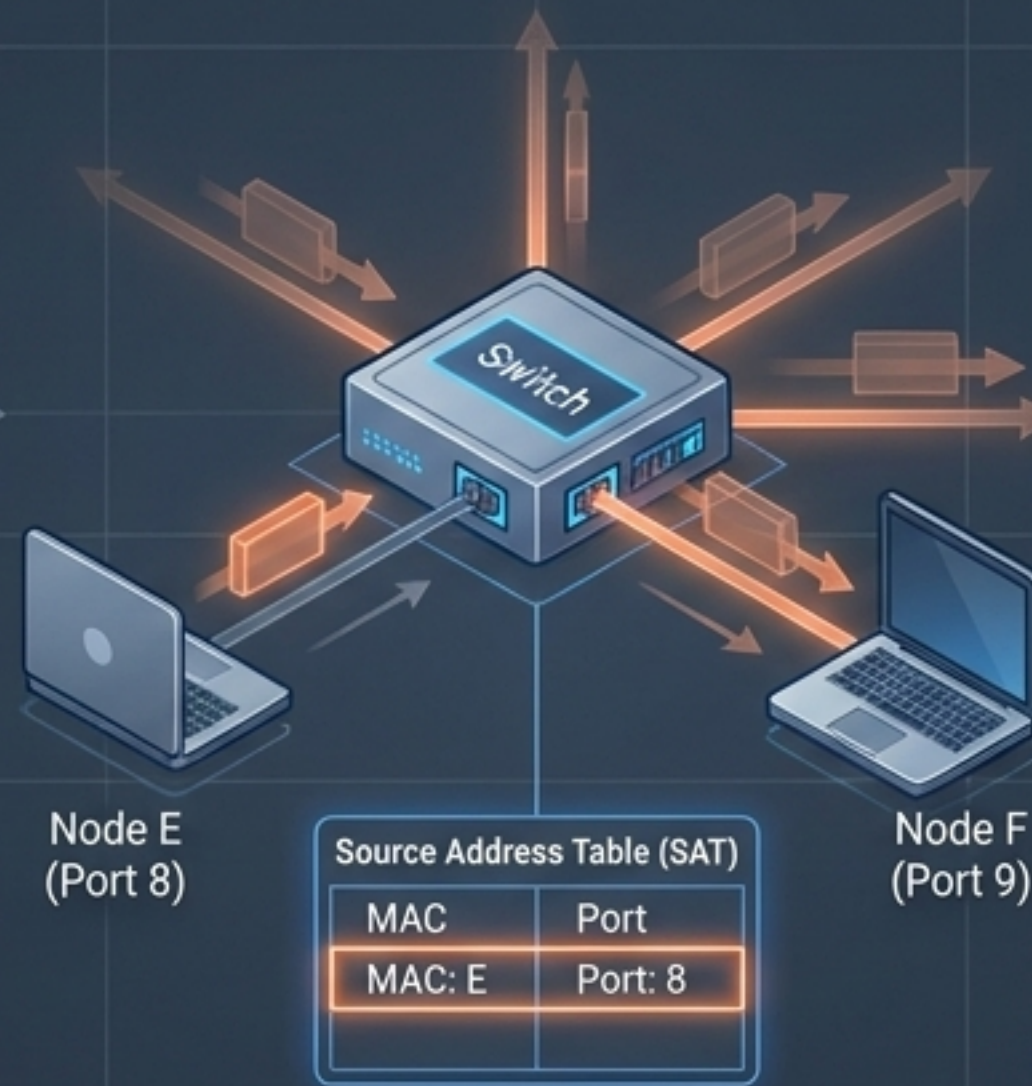
- **Scope:** Used by Routers to navigate across different networks (WAN).

Switches Dynamically Learn Network Layouts via the SAT

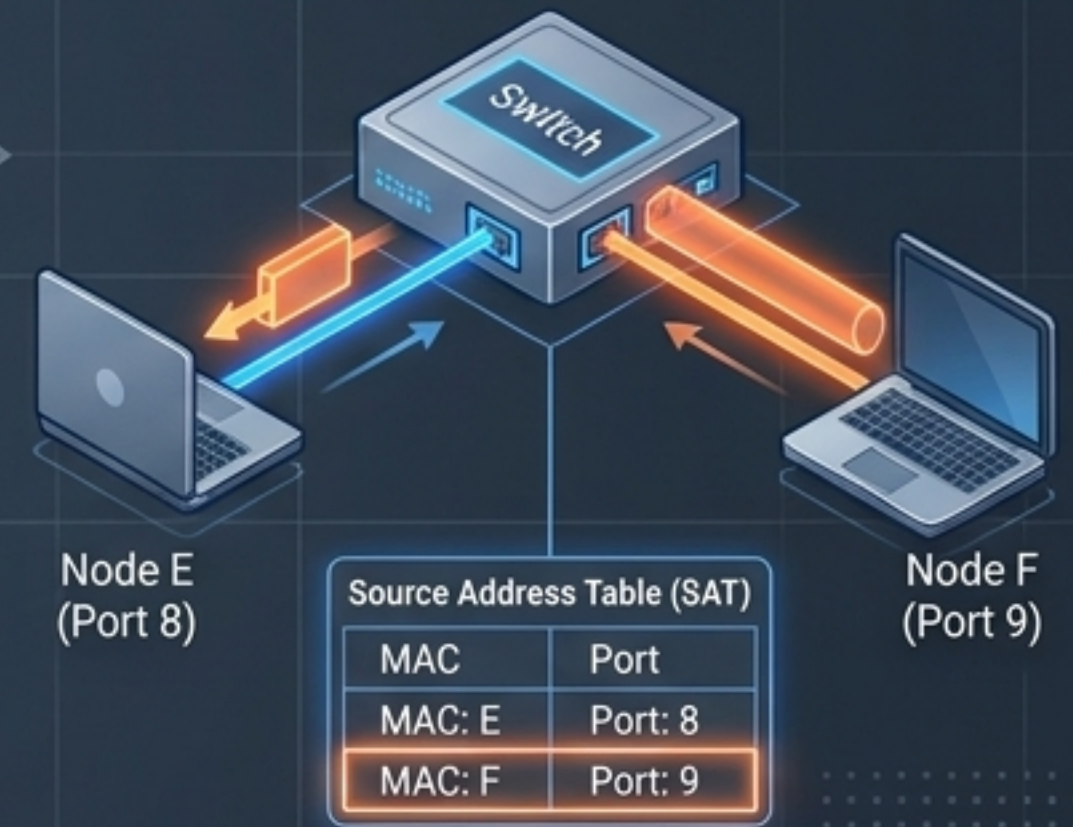
Step 1: The First Frame



Step 2: The Broadcast



Step 3: The Reply & Log



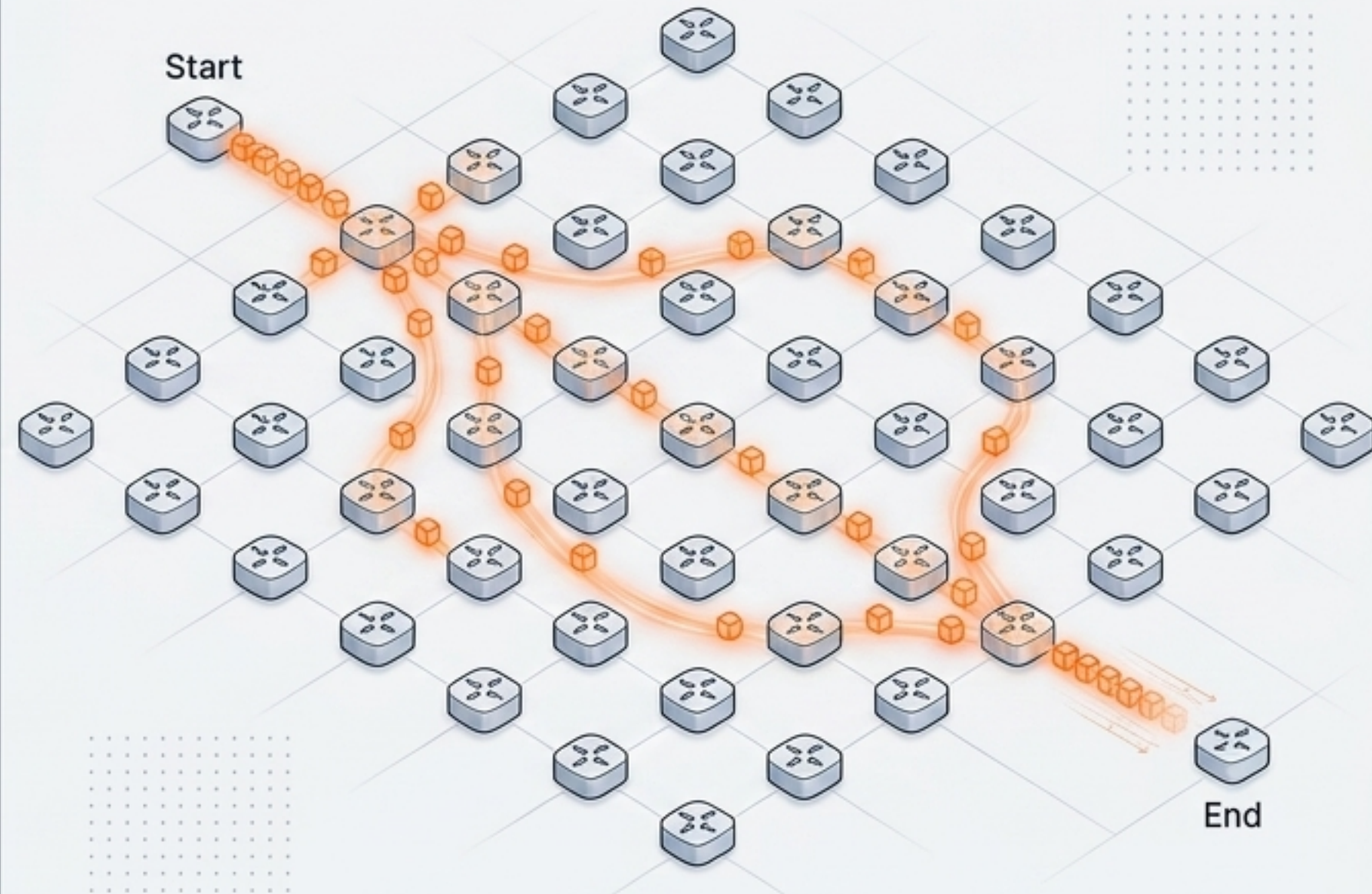
The Switching Matrix: Dedicated Lines vs. Dynamic Paths

Circuit Switching (The Old Standard)



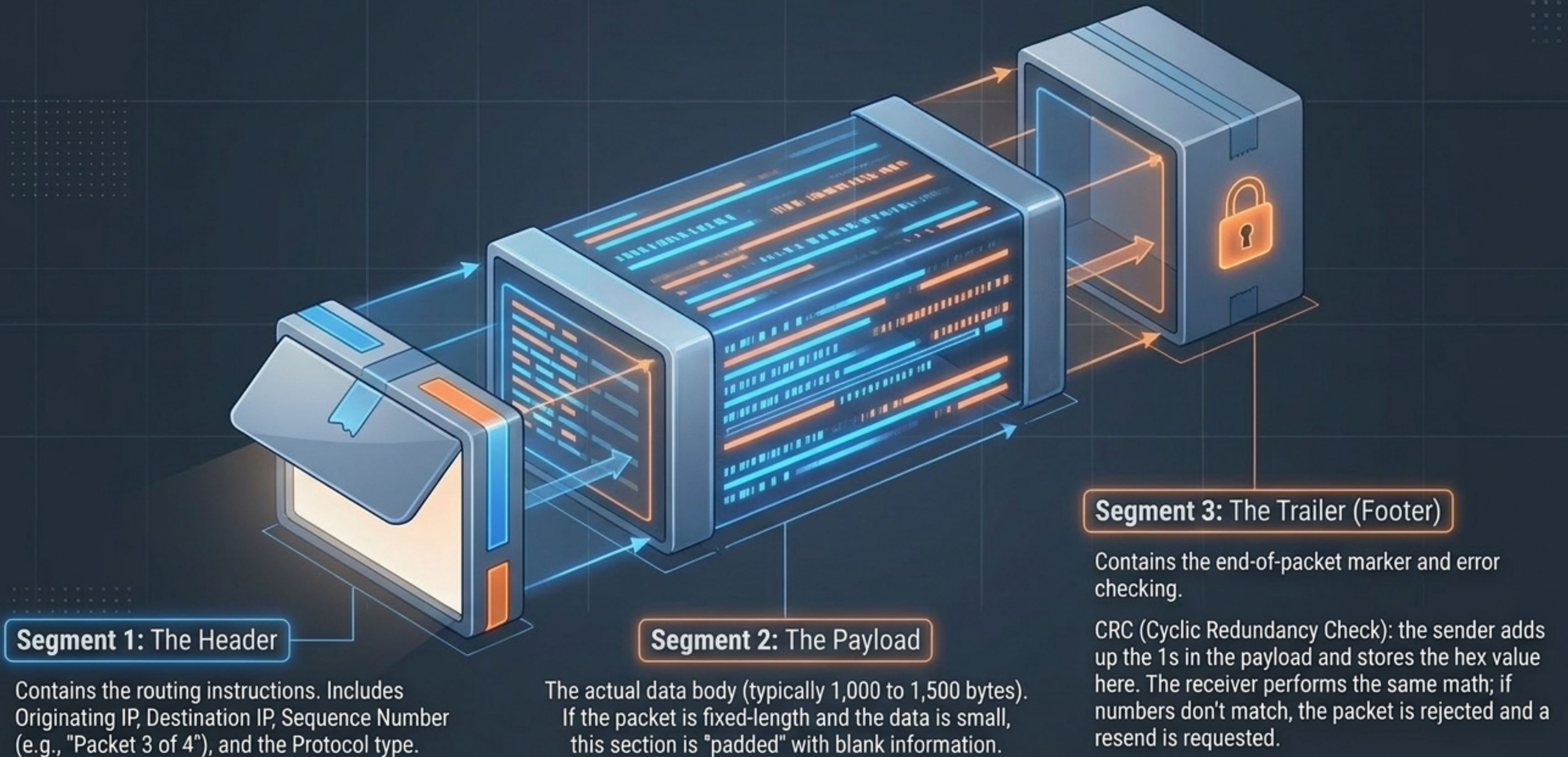
- **Mechanism:** A dedicated channel is established and held open for the duration of communication.
- **Use Case:** Traditional PSTN telephone calls. Guarantees bandwidth, but highly inefficient if data isn't constantly flowing.

Packet Switching (The Internet Standard)

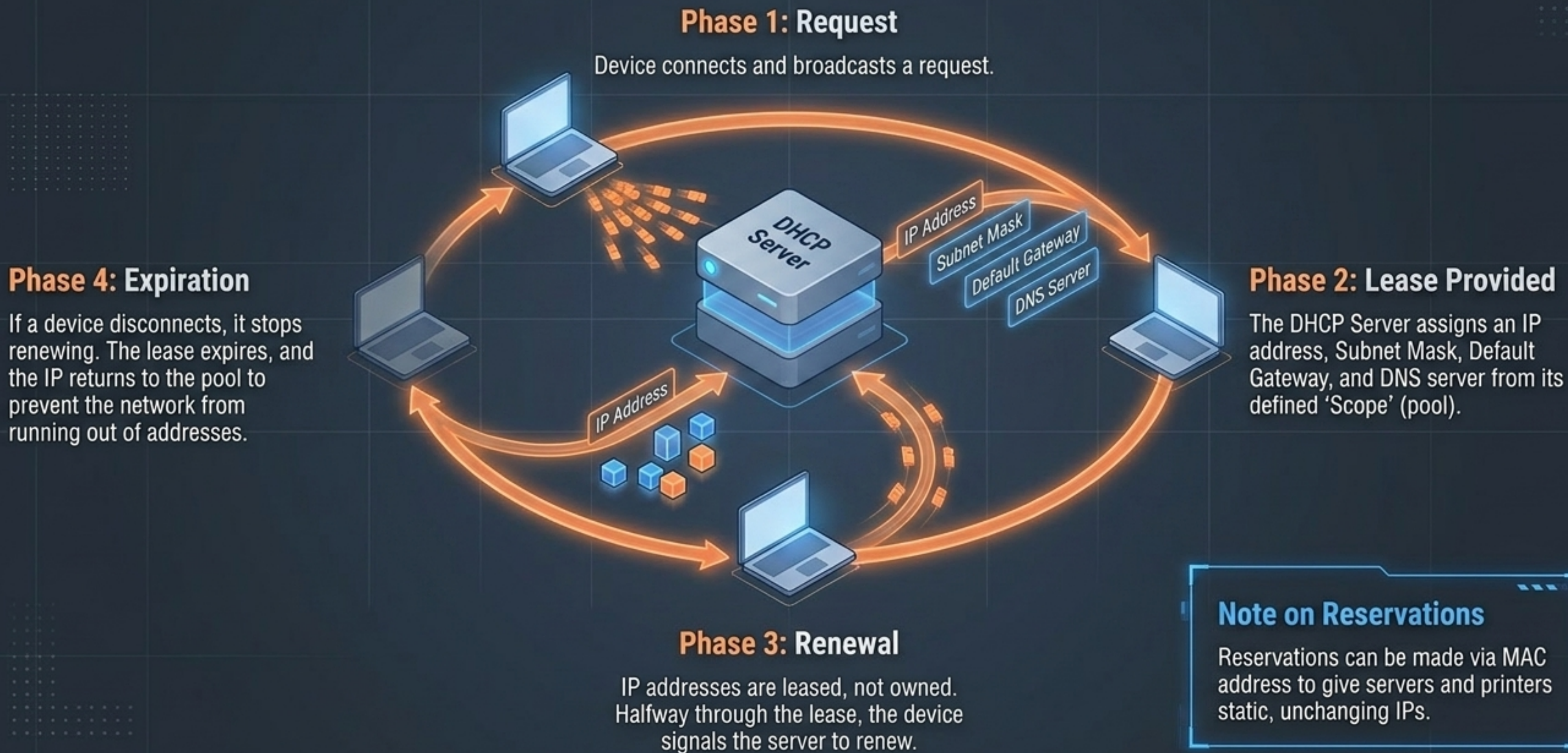


- **Mechanism:** Data is broken into packets. Each packet travels independently, finding the fastest available path. They arrive out of order and reassemble.
- **Use Case:** TCP/IP Digital Data. Highly efficient, routes around broken equipment instantly.

The Anatomy of a Data Packet



DHCP Automatically Manages the IP Address Pool



DNS Operates as the Internet's Layered Phonebook



The Email Protocol Diagnostic: SMTP, POP3, IMAP



SMTP (Simple Mail Transfer Protocol)

Function: Sending. ('Sending Mail To People').

Mechanism: Uses TCP to push outbound emails from the local client to the email server, and then to the recipient's server.



POP3 (Post Office Protocol 3)

Function: Downloading (Receiving).

Mechanism: Downloads the inbox to a single device and deletes the server copy. Saves server storage and allows offline viewing, but folder structures do not sync across devices.



IMAP (Internet Message Access Protocol)

Function: Synchronizing (Receiving).

Mechanism: Caches local copies but leaves the master file on the server. All folders (sent, drafts, custom) perfectly sync across laptops, phones, and tablets.

The Global Internet is a Deep-Sea Physical Infrastructure

The Backbone

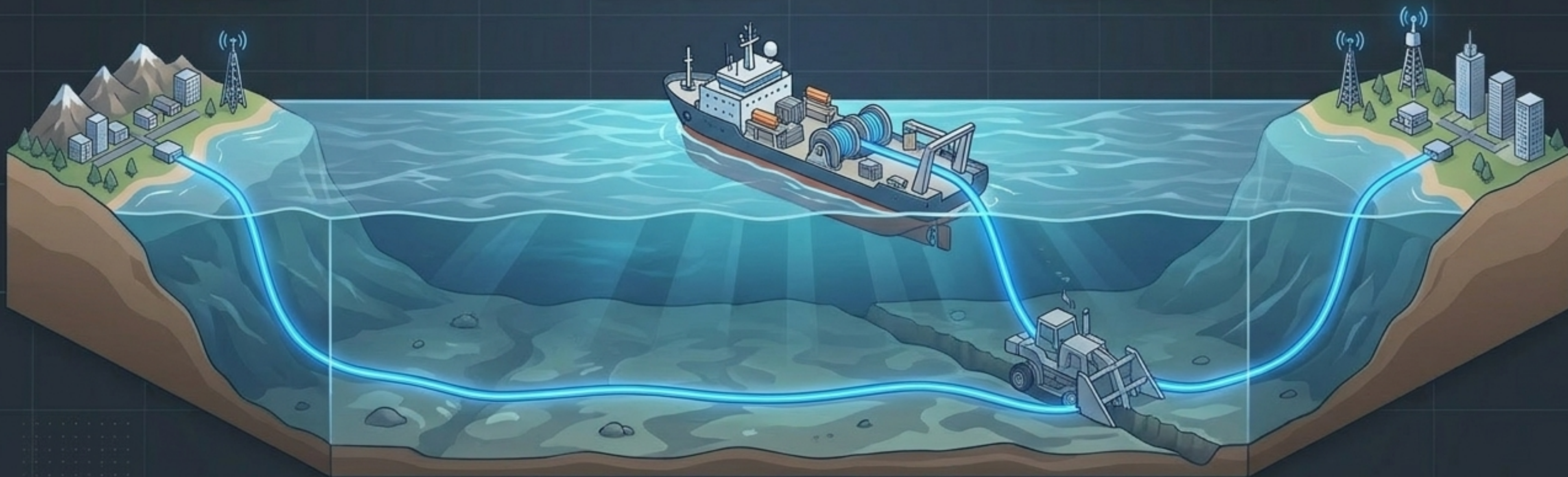
The internet is not a satellite cloud; it is a complex web of fiber-optic cables trenched into the seabed. Light pulses travel thousands of miles through these tubes.

The Transition

Cables reach the shoreline, connecting to massive routers that convert light pulses into electrical signals for local ISP distribution via copper or cellular towers.

The Governance

This massive global addressing and domain naming system requires standardization, managed centrally by ICANN (Internet Corporation for Assigned Names and Numbers).



Synthesis: The Anatomy of a Web Request

1. Client Request

A laptop connects to the LAN. DHCP leases it a local IP. The user types a URL.

2. Lookup

The request hits the DNS loop, resolving the human URL into a target IP Address.

4. The Gateway

The local Router reads the IP, realizes the target is external, and pushes it to the WAN.

5. Packet Switching

Data breaks down into Packets with Headers and Trailers, flying independently across subsea fiber optics.

6. The Server

Packets arrive at the destination Data Center Server, reassemble in order (checking the CRC), and the webpage is sent back.

